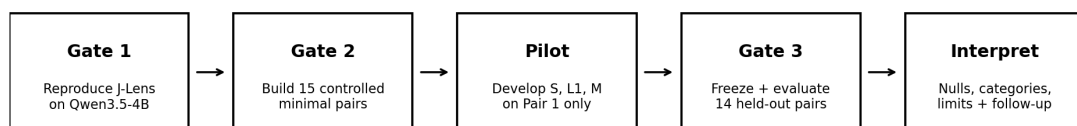


Finish My Sentence.

A Controlled Pilot Separating Activation-Derived Structure from First-Token-Conditioned Continuation in Multi-Token J-Lens

Pinar Patton, Model: Qwen/Qwen3.5-4B | 15 controlled minimal pairs | 1 pilot + 14 held-out | J-Lens reproduction + lightweight multi-token readout



One pilot pair set the ruler; the remaining 14 pairs were the test.

Figure 1: One pilot pair set the ruler; the remaining 14 pairs were the test.

Executive Summary

What I wanted to understand in this project was this: if a multi-token readout recovers a correct phrase, how much of that information is really represented in the activations, and how much of it is completed by the model's own continuation prior once it is given a useful first token? A multi-token output can look clean and convincing, but that alone doesn't tell you where the information came from.

I tested this on Qwen3.5-4B with an open-source J-Lens implementation. I first reproduced a known J-Lens sanity check, then built 15 structural minimal pairs across five different relation types. Pair 1 was used only as a pilot to develop the measurement design. I then froze all the settings and evaluated the remaining 14 pairs as a held-out set, without changing anything.

Two conditions did the real work in the comparison. L1 is an oracle-head continuation baseline: the model is given the source context and the candidate's correct first token, and a frozen language model scores only the remaining tokens. M is a lightweight activation-side alignment metric: it measures how closely the direction that separates two source contexts in residual-activation space aligns with the direction that separates the two multi-token concepts.

A mixed pattern emerged: both signals were present, but continuation was far more consistent. L1 tracked the expected structural reversal in the correct direction on all 14 held-out pairs. The matched mean absolute shift was 2.1646, about 15.8 times the value of the 20-permutation null baseline (0.1370). M was correct in direction on 9 of the 14 pairs. That 9/14 result was not significant on its own, so for M I used a 500-sample Monte Carlo permutation test. I repeatedly broke the correct source-concept pairing, recomputed the mean alignment, and used those 500 randomized results as the null distribution. The real matched mean alignment was +0.0420, while the null mean was -0.0021, giving an empirical one-sided $p \approx 0.002$. This showed that M was capturing more than random alignment and carried a pair-specific structural signal. Even so, the continuation baseline was clearly more consistent.

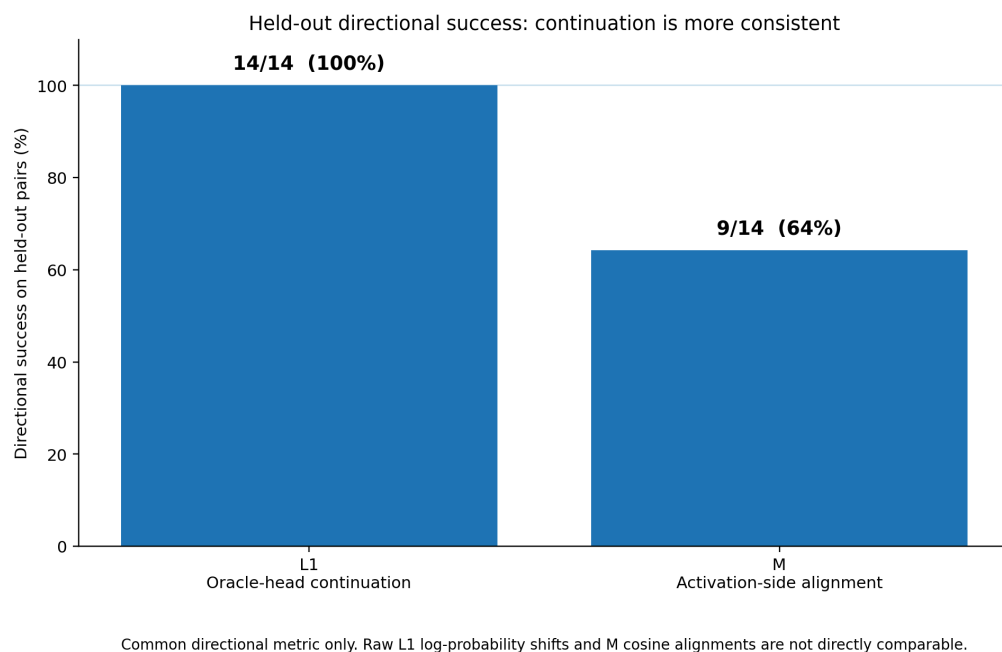


Figure 2: Held-out directional success on the common directional metric: L1 succeeded on 14/14 pairs, while M succeeded on 9/14. Raw magnitudes are not directly comparable because the metrics use different units.

Key Takeaways

- **First-token-conditioned continuation was the more consistent signal.** Given the correct first token, L1 captured the expected structural reversal in the right direction on 14/14 held-out pairs. The matched mean absolute shift was 2.1646. When I broke the source-candidate matching, this signal dropped sharply: about 23 times smaller under the mismatched-source control, and about 16 times smaller under the 20-permutation null. So I don't think what I'm seeing in L1

comes merely from the model's general phrase preference; it seems strongly tied to the correct source-candidate correspondence.

- **The activation-side signal was real, but less consistent.** M gave the correct direction on 9 of the 14 held-out pairs. But the 9/14 count alone was not significant, at $p \approx 0.15$. So I also ran a 500-sample Monte Carlo permutation test. The real matched mean alignment was +0.0420, versus a null mean of -0.0021 for the randomized pairings, giving an empirical $p \approx 0.002$. So the alignment I see in M doesn't seem to come from random pairing alone. Even so, there was no pair where M succeeded on its own.
- **Recoverability varied by relation type.** M didn't behave the same way across every relation type. Results were generally positive for role/argument binding, modifier attachment, and spatial relation. Quantifier binding showed no clear direction. Comparative relation was the most clearly problematic category: all three pairs had negative mean alignment. Post-hoc whitening moved the comparative mean from -0.0786 to -0.0391 and flipped Pair 11's sign, but the category still failed to clear its own null, $p = 1.0$. So whitening softened the result somewhat but didn't resolve the comparative failure.

What this tells me is that there really is a signal on the activation side, but the continuation side is far more consistent. In other words, both are present, they're just not equally strong or equally regular.

M and L1 are not two fully independent causal components. Continuation already arises from the model's own internal activations and representations. What's more, I'm not measuring the two with the same metric: L1 uses a log-probability shift, while M uses cosine alignment. Therefore, it wouldn't be valid to conclude with percentages like 70% continuation or 30% activation.

I think the real question this pilot raises involves readout provenance. A readout can produce the correct phrase without that meaning the whole phrase was read out directly from the activations. The model can complete the rest after the first token using its own continuation prior. So the distinction that matters to me is this: "the lens said the right thing" and "this information really came from the activations" are not the same claim. Separating exactly these two things is what I was trying to do in this project.